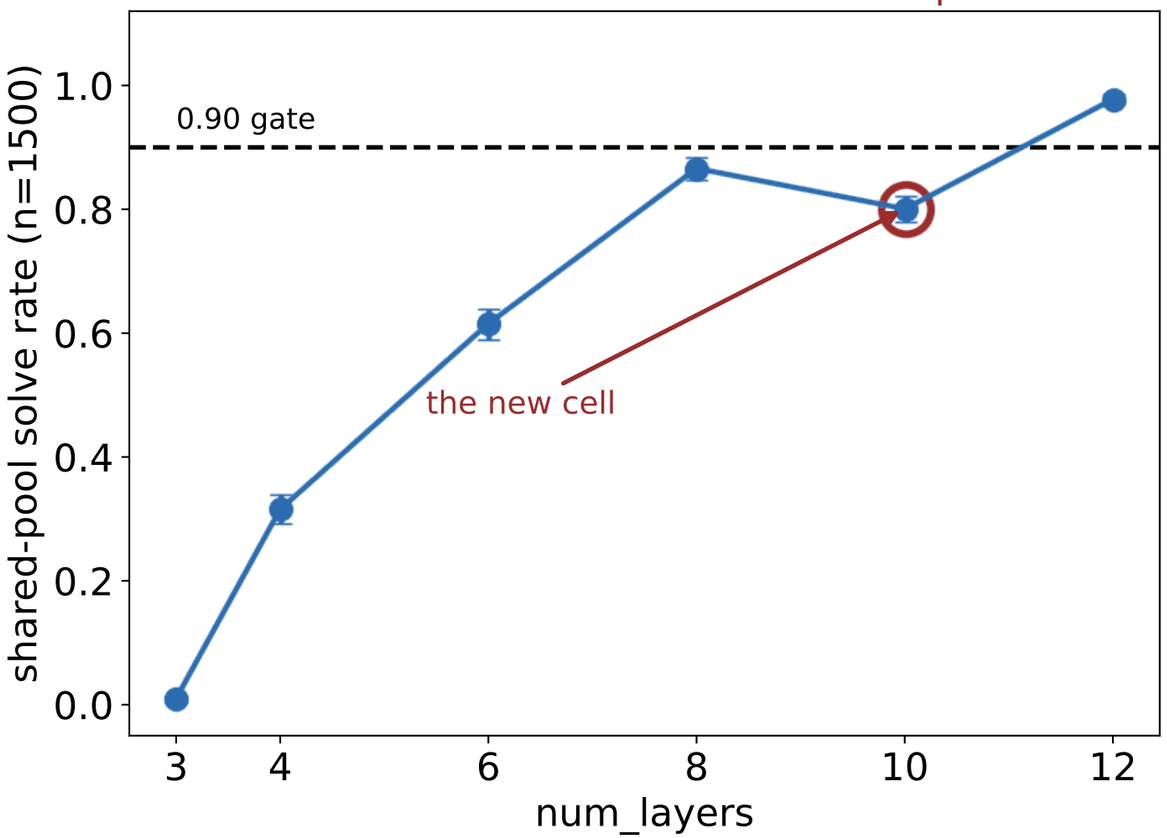
The midpoint cell that was supposed to halve the bracket

Finding 38 left L* in (8, 12] and said L=10 was the cheapest way to narrow it. It did not narrow.

L=10 came back at 0.8007 -- BELOW L=8's 0.8660.

paired on the same 1500 instances: -0.0653, discordant 92-190, McNemar exact p = 5.4e-09

the ladder is NOT monotone in depth




```
WHAT WAS PRE-REGISTERED
  pool in [0.93, 0.97]
  clears all 6 bins at rho=0.60
  L* narrows to (8, 10]

WHAT HAPPENED
  pool 0.8007
  clears 2 of 6 bins
  L* still 12; bracket still (8, 12]

WHAT IT ACTUALLY SHOWS
  not 'depth 10 is bad' --
  'you cannot read a knee off a
   single-seed ladder', because
   this one is not monotone and
   no rung has a seed replicate.
```

Job 50607549, commit 44f581f, seed 0, 8.03 M params, 8 h 59 m on one GPU. num_layers 10 is the only difference from L=12 and L=8, checked against each checkpoint's own config.json. L=6/4/3 are finding 33's cells, re-scored inside this job and identical instance for instance.

What the p-value does and does not buy

The result is sharp. Its interpretation is the part that needs care.

THE COMPARISON IS PAIRED AND CLEAN.

Both cells see the same 1500 held-out instances at the same seed_start, disjoint from training. The discordant split is 92-190 and McNemar exact gives p = 5.4e-09. Grammar is clean on both sides, so no part of the gap is formatting. Nothing about the measurement is in doubt.

BUT IT COMPARES CHECKPOINTS, NOT PROCEDURES.

What is established is that THESE TWO TRAINED MODELS differ by 0.065 on these instances. What is NOT established is that retraining L=10 would land below L=8 again. There is one seed per cell on this entire ladder and no rung has ever been replicated, so the p-value says nothing about the depth, only about the run.

TWO READINGS SURVIVE, AND ONE SEED CANNOT SEPARATE THEM.

H1, the leading one: optimization noise -- an unlucky trajectory at L=10. The final val_nll gap is 0.0026, about a thousandth, and it maps to 0.065 of solve rate, because free generation compounds per-token error over a whole trace; a noise-sized perturbation is therefore sufficient. H2, the interesting one: some depths genuinely train worse at this width and budget. H2 is the less likely and also unfalsifiable from a single seed.

CONSEQUENCE FOR EVERY L* IN THIS PROJECT.

L* is read as the first rung clearing 0.90 in all 6 bins, which assumes depth helps monotonically -- that a rung below the gate implies every shallower rung is too. This cell is a counterexample, so finding 38's bracket cannot be narrowed by interpolation and findings 11/14/22/33/38 need re-reading with error bars.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image:
node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 393 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_1 N2 COLOR_2 N3 COLOR_3 N5 COLOR_0 N8 COLOR_2 N9 COLOR_2 ... (59 tokens)
GRAPH	BEGIN_GRAPH N0 N10 N22 N7 N9 N44 N1 N39 N27 N23 N40 N21 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N1 COLOR_1 N4 COLOR_3 N7 COLOR_0 N16 COLOR_2 N17 COLOR_3 N18 ... (45 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 5 rounds, 19 deductions,
38 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000023.

One real reasoning trace

29 hints of 48 nodes, then 5 rounds of simultaneous deduction. Hints: N0=COLOR_1 N2=COLOR_2 N3=COLOR_3 N5=COLOR_0 N8=COLOR_2 N9=COLOR_2 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N1 COLOR_1 N4 COLOR_3 N7 COLOR_0 N16 COLOR_2 ...
  10 deduced
BEGIN_ROUND          N6 COLOR_0 N24 COLOR_0 N27 COLOR_3 N31 COLOR_0 ...
  5 deduced
BEGIN_ROUND          N26 COLOR_2 N41 COLOR_1
  2 deduced
BEGIN_ROUND          N10 COLOR_3
  1 deduced
... 1 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N1, neighbours N39 N27 N23 N40 N21

their colours known before this round:

```
   [0, 2, 3]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N27
```

only colour left: 1

```
so the trace emits  N1 COLOR_1
```

The model is never shown this reasoning -- only the node/colour pair.

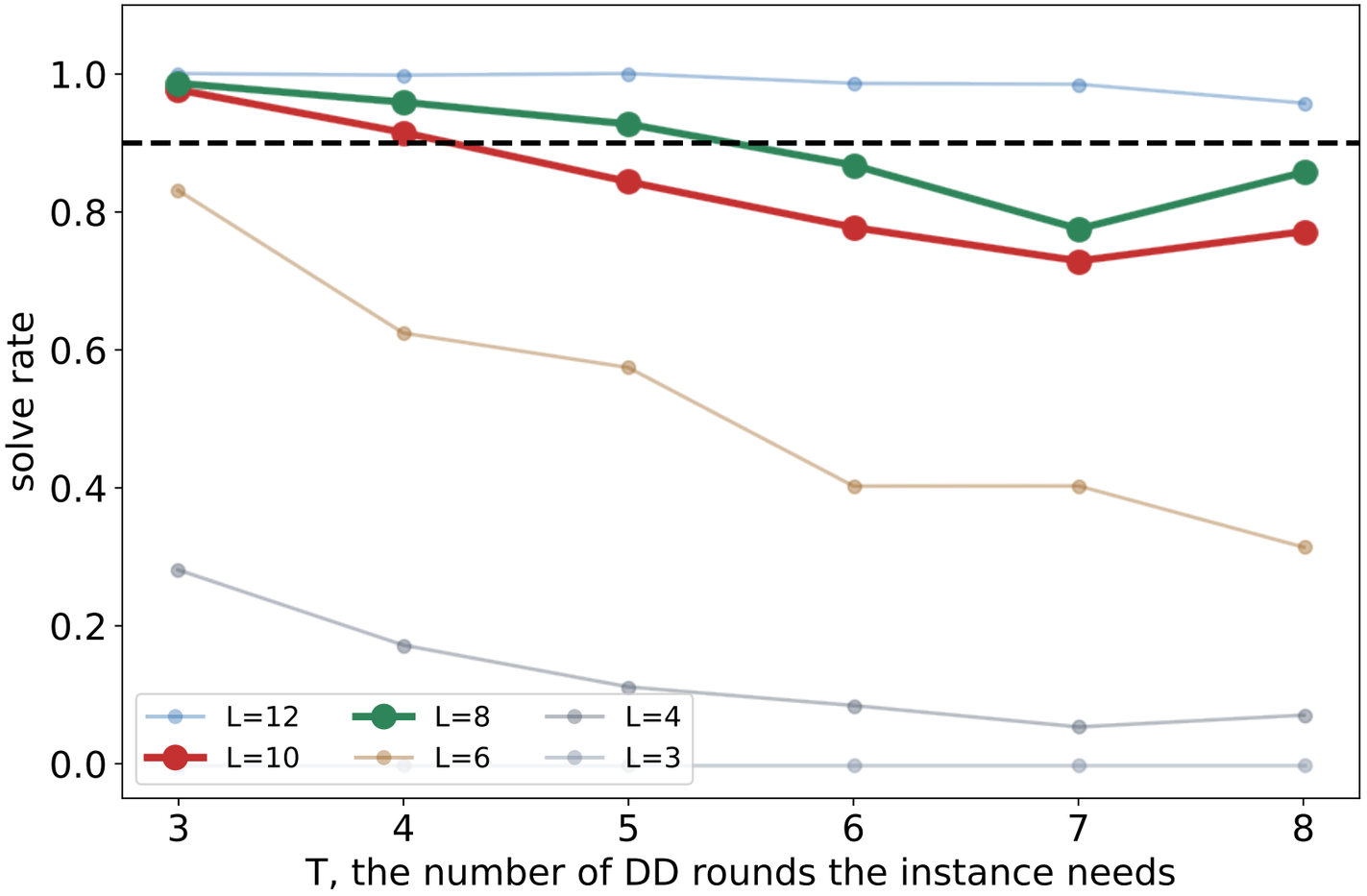
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

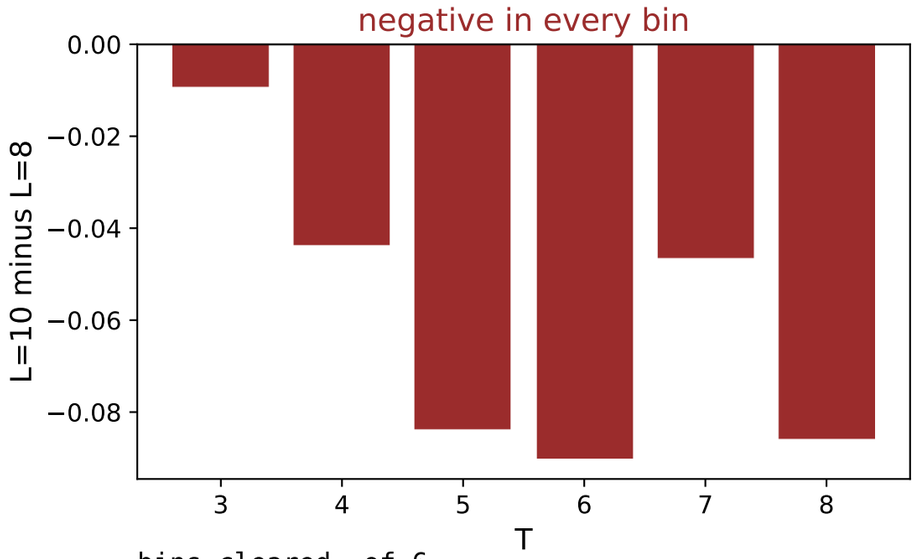
T is the x-axis of every plot in this deck: it is the number of DD ROUNDS this instance needs, counted on the trace above -- not the number of tokens and not the number of nodes. The depth bins group held-out instances by exactly this number, at fixed density, and L* asks how many layers the network needs before it clears 0.90 in all of them. The cell on the previous pages is the one that was supposed to sit between L=8 and L=12 on that scale.

T=5 rounds, the 90th percentile of 40 draws from this config (range 1-9) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Bin by bin at density 0.60: L=10 sits under L=8 everywhere

Bins group held-out instances by T, the DD rounds they need. A cell owns L* only if its whole line clears the dashed gate.






```
bins cleared, of 6

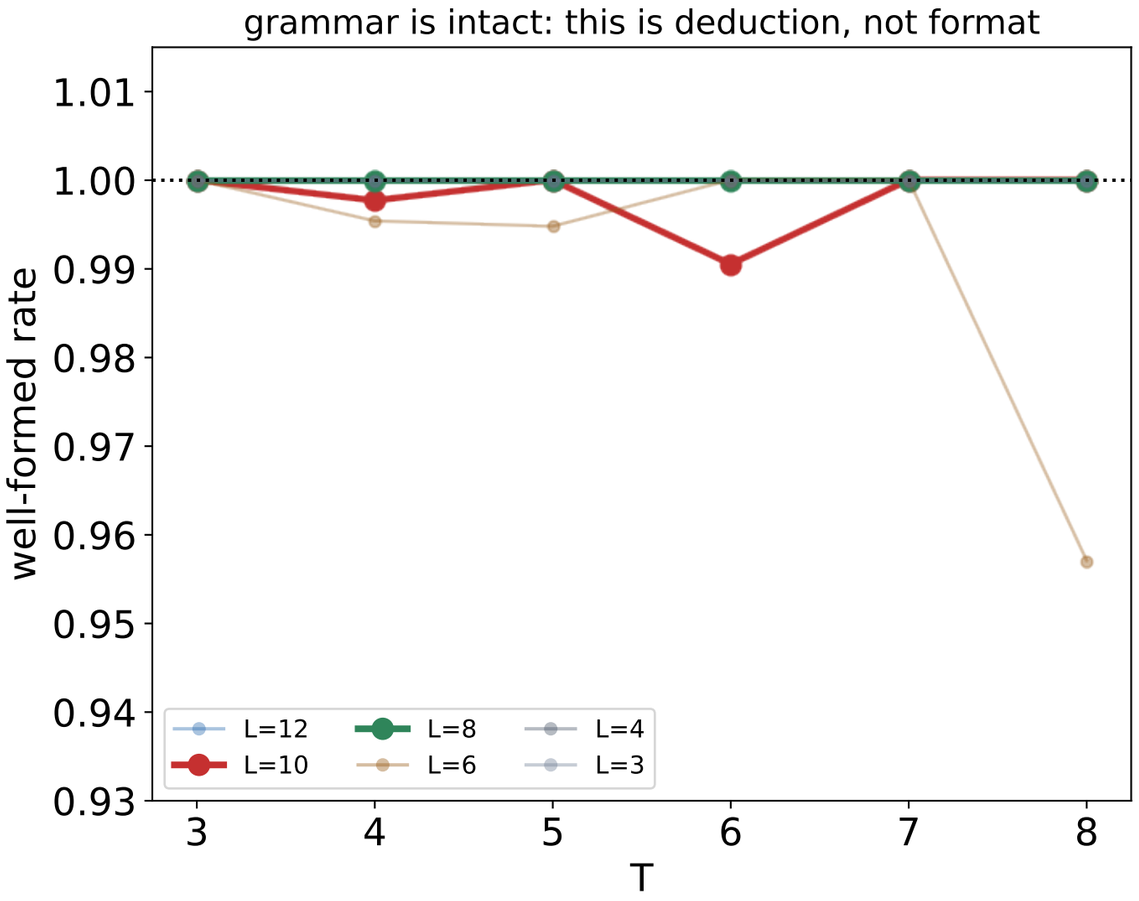
  L=12  6/6  <-- clears, owns L*
  L=10  2/6
  L=8   3/6
  L=6   0/6
  L=4   0/6
  L=3   0/6

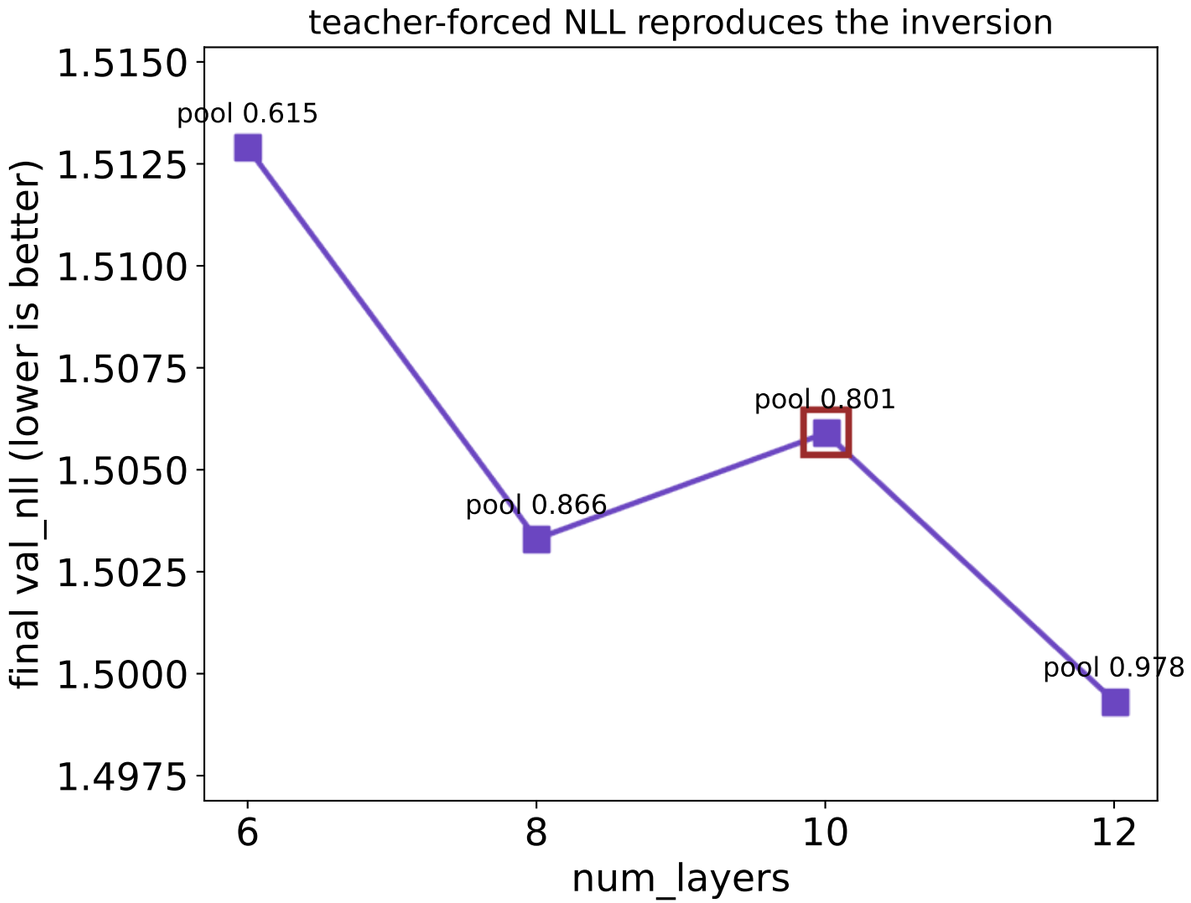
density 0.5834-0.5839, spread 0.000488
  the control: deeper is not sparser
```

L=10 clears only T=3, 4; the four it misses are the four deepest and contiguous at the top -- the one pre-registration the result confirmed. Cochran-Armitage trend in T: L=10 z = -8.09, L=8 z = -7.65. The per-bin deltas are not monotone in T, and the three deepest bins hold only 211, 129, 70 instances, so only the overall sign is resolvable.

Is it the eval, or is it the model?

Two independent checks, and they agree: the deficit is already in the trained weights.

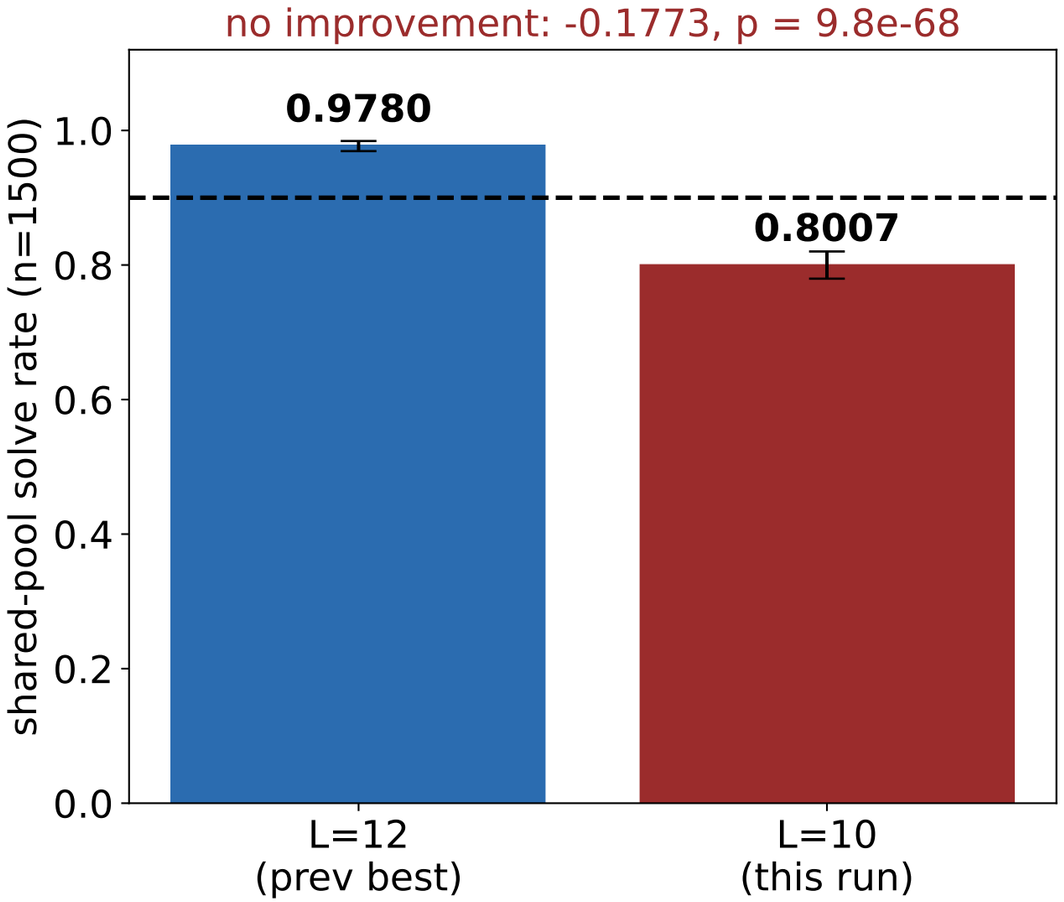




L=10 pool controls: truncated_rate 0.0007, outcome_accuracy 0.9953, minimum well-formed rate over all bins 0.9905 -- finding 25's grammar-collapse mechanism is excluded. Token accuracy ranks the rungs the same way: L=12 0.4967, L=10 0.4942, L=8 0.4947, L=6 0.4921. That is concordance, not a rehabilitation -- val_nll stays disqualified as a cross-model ranking statistic (findings 4/20/25, r = +0.33 over 34 runs) -- but it does put the gap before free generation.

Against the previous best on the same eval

Same 1500 instances, same protocol, so this is a paired test rather than a comparison of two averages.



(a) HEADLINE FALSIFIED

'L=10 clears 0.90 in ALL SIX rho=0.60 bins, so L* narrows to (8, 10]' -> clears 2 of 6.

(b) POOL FALSIFIED

'in [0.93, 0.97], resolvably above L=8' -> 0.8007, below falsifier B (0.9220) AND below L=8. The stated prior said a miss above 0.97 was likelier; it missed low by 0.12. This is the project's first interval prediction to miss on the low side.

(c) CONDITIONAL CONFIRMED

scored because (a) failed: 'the missed bins are the deepest and contiguous' -> missed T=5,6,7,8, a contiguous run at the top. Its falsifier did not occur.

(d) CONTROLS CONFIRMED

realized density 0.583432-0.583920, within 0.001 of 0.5836; all repeat evaluations of the five reference cells identical instance for instance.

2 of 4. The two that failed are the two that predicted the result.

Pre-registrations were committed in the sbatch header (44f581f) before the job was submitted and are quoted from it verbatim; the analysis script was written while the job was at epoch 76 of 100. The best N=48 leak-free cell remains finding 38's L=12 at 0.9780. This run's value is the negative result, not a new top line.

The next job buys seeds, not rungs

Hypothesis first: retraining L=10 under a second seed lands ABOVE L=8, i.e. the inversion was noise (H1).

THE EXPERIMENT.

L=10 and L=8 retrained at seed 1, everything else identical -- same 240000 x 100 epochs, same ff=1024, same pool. Two cells, about 9 h and 7 h on one GPU each, and they can run concurrently. If both seeds put L=10 below L=8 the ladder really is non-monotone (H2) and every L* in this project needs re-reading. If seed 1 puts L=10 between L=8 and L=12, this cell was noise -- but the measured seed spread then becomes a required error bar on every future rung, which the ladder has never had.

WHY IT BLOCKS THE HYPER-PARAMETER SCAN.

The planned scan asks whether required reasoning depth tracks MLP depth, and this cell was going to be its centre. A scan whose rungs carry an unmeasured seed spread of at least 0.065 solve rate cannot see an effect smaller than that, and the depth effects it is looking for are plausibly that size. Measuring the spread first is cheaper than running the scan twice.

WHAT DOES NOT CHANGE.

L=12 still clears all six bins at 0.9780 and L=6 still clears none, so the (8, 12] bracket from finding 38 stands. The leak-free task at N=48 still needs far more depth than the leaked version did. Nothing here reopens finding 16.

LIMITATIONS.

One seed per cell, which is the headline rather than a caveat. L=9 and L=11 are unmeasured, so the bracket holds four integers of which two are known. Everything is at ff=1024 (finding 14) and in the fast phase above the removal floor (finding 34). The job's wall clock spans the DST fall-back; Elapsed 8:58:51 is real time.

Negative result, reported as one: 8 h 59 m of GPU bought a cell that cannot be interpreted without a replicate. The methodological finding is worth more than the number.